

Cross-Domain Authentication Scheme for IoT Devices Based on Blockchain

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Abstract—With the popularity of Internet of Things (IoT) technology, the scale of IoT is gradually becoming huge and demand-driven to form multi-domain IoT, and the demand for cross-domain access of devices in multi-domain scenarios has greatly increased. Secure cross-domain authentication is an important step in securing multi-domain IoT. The current cross-domain authentication architecture for IoT devices has limitations such as single point of failure, low authentication efficiency, poor scalability and other limitations. To address these problems, this paper designs a blockchain-based cross-domain authentication scheme for IoT terminals, which adopts an authentication architecture based on a combination of blockchain and edge computing technology to eliminate the problems of traditional cross-domain authentication for IoT devices and achieve distributed authentication. Considering that IoT devices have the characteristics of heterogeneity, massiveness, and limited resources, a lightweight and secure certificateless signature scheme is introduced in the cross-domain authentication process, eliminating complex certificate management and key escrow, and achieving secure and efficient cross-domain authentication of IoT terminals.

Keywords—Blockchain; IoT; Cross-domain; Authentication.

I. INTRODUCTION

With the rapid development of Internet of Things (IoT) technology, the application of IoT can be found everywhere in daily life and play an important role in various fields [1]. As the access devices of the Internet of Things, the terminal devices are the source of IoT data and are an important part of IoT. IoT devices often involve multiple management domains, and the cross-domain access requirements of each management domain in the IoT environment are greatly increased. IoT application scenarios such as Industrial Internet, Telematics, Smart Home, Smart City and Smart Healthcare usually involve several different management domains. In these scenarios, more and more devices will constantly move from one domain to another, devices in one domain need to access networks or data in other domains, and devices in different domains need to collaborate with each other to complete their tasks. In the Industrial Internet scenario, the whole manufacturing process usually involves several different management domains, with devices in different domains collaborating with each other to complete production tasks. In the Smart Home scenario, IoT

devices in different management domains exchange data to make more rational decisions and improve the efficiency of tasks. In the healthcare scenario, doctors in different hospitals have to access and share patients' medical data in order to make better decisions and propose the correct and timely treatment plan for the patient. In order to prevent attacks launched by devices with unknown identities from posing security threats to the Internet of Things, cross-domain identity authentication is required for secure communication between devices from different domains. The core problem of cross-domain authentication is how one domain recognizes the identities of devices from another domain. Traditional authentication schemes mostly rely on a trusted third party in scenarios where a large number of IoT terminals have cross-domain access [2-4], which not only has problems such as high operation cost and low authentication efficiency, but also has the hidden danger of centralization, and the security factor will be greatly reduced. Therefore, how to achieve secure and efficient cross-domain authentication of devices and ensure the security of the IoT is a hot research topic nowadays.

In recent years, blockchain technology received extensive attention in identity authentication research due to its decentralised nature [5]. In the literature [6], a distributed secure access scheme for IoT based on blockchain technology is proposed for the traditional access authentication scheme which mainly relies on digital certificates for authentication and has the problem of high communication overhead, but this scheme does not achieve cross-domain authentication. Blockchain is a distributed ledger maintained by multiple cooperative peer nodes. Nodes must be verified before joining the network and there is no complete trust between nodes. Therefore, the blockchain can be used to build trust between different domains. Zhou et al [7] proposed an efficient cross-domain authentication scheme based on blockchain technology by setting the authentication server and root CA certificate server of each domain as blockchain nodes based on the PKI authentication system. The literature [8] uses federated blockchain technology to construct a decentralised network with a root certificate authority as the authentication node. The hash value of the authorization certificate is stored in each block. The verification process only needs to compare whether the hash value is consistent, which improves the authentication

efficiency. Edge computing is a new computing model for computing at the edge of the network [9], which can process requests from massive terminal devices in real time. Applying edge computing to cross-domain authentication and using authentication servers at the edge of the network to decentralise the processing of authentication requests can solve the authentication latency problem, reduce the network burden and achieve efficient cross-domain authentication. Zhang et al [10]. proposed a blockchain based cross domain authentication and key agreement protocol in the edge computing environment for cross domain authentication and key agreement of low performance IoT devices, built an alliance blockchain under the "edge-end" architecture, and stored the hash value of the terminal device certificate in the blockchain, reducing the authentication cost and improving the authentication reliability. As a distributed ledger, the blockchain technology has a natural match with the decentralized edge computing model. The blockchain technology provides a trusted and secure environment for edge computing network services [11]. The literature [12] proposes a lightweight secure edge computing architecture based on blockchain, which realizes the organic integration of edge computing and blockchain. Based on the above research, this paper combines the blockchain and edge computing, and adopts a cross domain identity authentication architecture of IoT devices based on the main-slave blockchain, which can realize the decentralization of IoT cross-domain authentication and the distributed management of the cross-domain authentication process, solve the single point of failure in authentication, reduce the burden of blockchain nodes, improve the efficiency of cross-domain authentication, and guarantee the secure communication of devices in different domains.

IoT devices often have the characteristics of heterogeneity, magnanimity and resource limitation, which brings certain challenges to the security of IoT. Digital signature technology is suitable for multi-domain IoT environment, and can provide identity authentication for resource constrained IoT devices. In the signature scheme, the traditional public key cryptosystem has the problem of complex certificate management. Although the identity based cryptosystem eliminates the certificate management, it has the problem of key escrow, which is easy to leak the user's privacy. In the process of cross domain identity authentication, this paper uses a lightweight and secure certificateless signature algorithm that does not introduce third parties to achieve cross domain identity authentication. It does not need to use trusted third party entities to issue certificates. It also eliminates the key escrow problem of identity based cryptosystems [13].

II. KEY TECHNOLOGY

A. Edge Computing

Edge computing is a new computing model. It provides converged computing, storage, and network resources for applications at the edge of the network near objects or data sources. Meanwhile, edge computing is also an enabling technology. It has the advantages of real-time data processing and analysis, high security, privacy protection, strong scalability, location awareness and low traffic [14-15]. Edge

computing is a lightweight computing that distributes information storage from the unified cloud to each edge, extract feature data and send it back to the cloud after intelligent processing on the edge side, which greatly reduces the storage and processing pressure on the cloud. Its core idea is to provide computing, storage, network and other resources for applications on the edge of the network. Edge computing architecture can meet the needs of users for delay sensitive applications and reduce the load pressure on the core network.

B. BlockChain

Blockchain was first proposed by Satoshi Nakamoto in his paper. Blockchain has properties such as decentralisation, tamper resistance, difficult to forge, and traceability. A blockchain is a distributed ledger involving multiple nodes, each of which stores complete data and does not rely on a central authority. Each node is independently peer-to-peer and the data is maintained by them together. A blockchain is made up of blocks that hold certain information. The blocks are connected in a chain structure in the chronological order in which they are generated, and cryptographic knowledge is used to guarantee their immutability and unforgeability [16-17]. The block header contains mainly the version number, hash value of the previous block, timestamp, random number, target hash, and MerkleRoot. The transaction ledger information recorded in the block body is based on the Merkle tree structure. Any change in the internal transaction data will cause the change of the transaction hash value, resulting in the change of the MerkleRoot hash value. Therefore, it can ensure that the data is not easily tampered with, difficult to forge, and traceable.

C. Digital Signature Technology

Digital signature is a digital string generated by the sender of a message, which can not be forged by others. It is a valid proof of the authenticity of the message sent by the sender. Digital signature technology [18] is a commonly used authentication technology of the Internet of Things. It is a specific application technology combining message digest function and public key encryption algorithm. In the digital signature technology, the signer is required to use his own private key to generate a signature for the information digest. The signer's private key is confidential, the public key is public, and the verifier needs the signer's public key to verify the signature. The information generated by digital signature technology has data integrity protection and non-repudiation. By verifying the signature to verify the identity of the signer, it can provide authentication for resource-constrained IoT devices and is suitable for the IoT environment.

III. OVERALL ARCHITECTURE OF THE SCHEME

The overall architecture of the scheme is shown in Figure 1: In the multi-domain IoT scenario, there are multiple domains, each of which contains:

(1) Registration Center (RS): generates part of the private keys for edge authentication nodes and terminal devices, and is responsible for the registration of edge authentication nodes and terminal devices in the domain.

(2) Edge Authentication Nodes (EA): scattered deployment at the edge of the network, close to the device side, and has strong computing power. It is used to authenticate the identity of terminal devices, reduce the authentication delay of the network, and improve the authentication efficiency.

(3) Terminal equipment (TD): the sensing layer of the Internet of Things, used for sensing, service and communication, is the source of data of the Internet of Things and an important part of the Internet of Things.

Each domain consists of an edge authentication node and a registry to form a slave chain. Each slave chain stores the device information of the domain, eliminating the hidden dangers of centralized authentication and realizing distributed authentication. The main chain stores the index of device information of each domain, which is composed of the registry of each domain as the node of the main chain. The main chain connects each slave chain through them to realize the sharing of device information. When a device accesses the local domain, it initiates an authentication request to the nearest edge authentication server, and the edge authentication node responds to the request in time to verify the identity of the terminal device by obtaining the public parameters of the end device and the registration information from the blockchain. When a device accesses across domain, the identity of the device is verified through the index information stored on the main chain to achieve cross-domain authentication. Different domains can ensure independence in different slave chains, and each domain does not need to store unnecessary authentication information, reducing the storage burden of the blockchain.

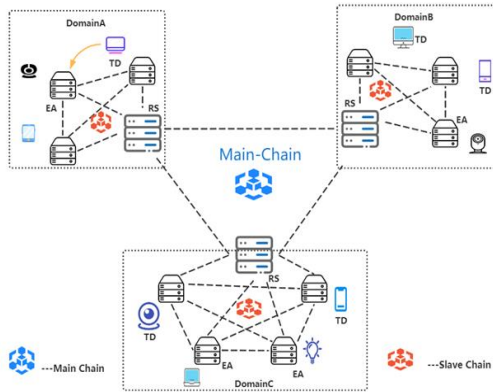


Figure 1. Overall Architecture of the Scheme

IV. PROCESS OF THE SCHEME

The overall process of this scheme is divided into four phases, including the initialization phase, the registration phase, the intra-domain authentication phase, and the cross-domain authentication phase. This scheme uses a signature scheme based on certificateless cryptography in the authentication process. The certificate-less signature scheme does not use public key certificates, which not only eliminates the need to use trusted third party entities to issue certificates, but also eliminates the key escrow issues of identity-based cryptosystems. At the same level of security, elliptic curve

cryptography [19] has the advantages of smaller private key size, less computation, smaller storage and smaller bandwidth. Therefore, this paper uses a certificateless signature without bilinear pairs based on elliptic curves to improve the authentication efficiency. The specific process as follow:

A. Initialization phase.

In the initialization phase, the registry of each domain generates system public parameters, system master key, generates its own key pair, generates partial private keys for edge computing nodes and end devices, and stores the relevant information of this domain in the blockchain, and the devices generate their own public-private key pairs based on the partial private keys.

1) Initialization of registration center.

The KGC of the registration center picks the security parameter l as input and picks the elliptic curve $y^2 = x^3 + b \pmod{q}$ over the selected finite field, where $x, y \in [0, q-1]$ and $4a^3 + 27b^2 \pmod{q} \neq 0$. The notation q is a large prime with a length of l bits, where l is a secure parameter. G is an elliptic curve additive group with order q , and P is a generating element of G . Randomly select a value $x \in \mathbb{Z}_q^*$, calculate $P_{pub} = xP$, and secretly save x as the system master key. Select three different secure hash functions: $H1: \{0, 1\}^* \rightarrow \mathbb{Z}_q^*$, $H2: \{0, 1\}^* \rightarrow \mathbb{Z}_q^*$, $H3: \{0, 1\}^* \rightarrow \mathbb{Z}_q^*$. Then the registry generates asymmetric key pairs (RS_{SK}, RS_{PK}) , publish params: $\{l, G, q, P, P_{pub}, H1, H2, H3\}$ and RS_{PK} , and stores $(RS_{ID}, params, RS_{PK})$ in the slave chain and its index information $(RS_{ID}, block_num)$ in the main chain.

2) Initialization of devices.

The KGC of the registration center is randomly selected $\alpha \in \mathbb{Z}_q^*$, and calculates $y_{ID} = \alpha P$, $h1 = H1(ID, y_{ID}, P, P_{pub})$, $d_{ID} = \alpha + h1 \cdot x \pmod{q}$. It generates partial private key $S_{ID} = (y_{ID}, d_{ID})$, and return it to the device secretly. After the device receives the partial private key, it randomly selects $v_{ID} \in \mathbb{Z}_q^*$ as the secret value and computes $g_{ID} = v_{ID}P$. $PK_{ID} = (g_{ID}, y_{ID})$ is the public key of the device ID and the private key is $SK_{ID} = (v_{ID}, d_{ID})$.

B. Registration phase.

In this phase, it contains the registration of IoT devices and edge authentication nodes. The IoT device registers with the domain registry, and upon successful registration, the device is associated with the domain, and the information about the device is stored in the slave chain and the related index information is stored in the main chain.

1) Registration of IoT devices.

a) TD $\rightarrow$ RS:

$\{RS_{ID} || TD_{ID} || TD_{PK} || R1 || LT || T || Sig_{TD}(TD_{ID} || RS_{ID} || R1 || LT || T)\}$
The device sends a registration request to the registry, where $R1$ represents the registration request, LT represents the life cycle, T represents the timestamp, and $Sig_{TD}(*)$ represents the

signature generated by the device TD invoking the signature algorithm (Algorithm 1) using the private key.

b) After receiving the message, RS first checks whether the RS_{ID} sent by the device is correct. If the RS_{ID} is not correct, it indicates that it is not the current domain, and a failure message is returned. Otherwise, RS checks whether TD_{ID} already exists in the slave chain. If it exists, it indicates that the device has been registered, and a failure message is returned. If it does not exist, continue to verify. Invoke the verify algorithm (Algorithm 2) and use the public key of the device to verify the validity of the device signature. If the verification is successful, proceed to the next step. Otherwise, the registration is terminated.

c) $RS \rightarrow BC_s : \{TD_{ID} || RS_{ID} || LT || T || Card\}$. RS uses the private key to sign ($TD_{ID} || RS_{ID} || LT || T$) by calling the signature algorithm (Algorithm 1), authorizes the device, generates the identity credential: $Card = Sig_{RS}(TD_{ID} || RS_{ID} || LT)$, and then deposits the: $\{TD_{ID} || RS_{ID} || LT || T || Card\}$ into the slave chain.

d) $RS \rightarrow BC_m : \{TD_{ID}, RS_{ID}, BC_s_num\}$. RS deposits TD_{ID}, RS_{ID} and the block number of the corresponding slave chain into the main chain.

e) $RS \rightarrow TD : \{Card\}$. RS sends Card to the device.

f) The device verifies the authenticity of the registry identity by verifying the signature and stores the identity credential.

2) Registration of edge authentication node.

The registration of edge authentication nodes is the same as the registration process of IoT devices, so we will not go over it here.

Algorithm 1 Signature-Algorithm

Input: message m , S_{ID} , v_{ID} , and $PK_{ID} = (g_{ID}, y_{ID})$.

Output: signature $\sigma = (\delta, z)$.

1. Randomly pick $k \in \mathbb{Z}_q^*$ and compute $\delta = kP$.
 2. Compute $h2 = H2(m, \delta, ID, PK_{ID}, P_{pub})$,
 $h3 = H3(m, \delta, ID, PK_{ID}, h2)$.
 3. Compute $z = k^{-1}(h2v_{ID} + h3d_{ID}) \mod q$.
 4. The signature on message m is $\sigma = (\delta, z)$.
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Algorithm 2 Verify-Algorithm

Input: a message-signature tuple $(m, \sigma = (\delta, z))$, params, and $PK_{ID} = (g_{ID}, y_{ID})$.

Output: verification result—1 or 0.

1. Calculate $h1 = H1(ID, y_{ID}, P, P_{pub})$,

$h2 = H2(m, \delta, ID, PK_{ID}, P_{pub})$.

2. Compute $h3 = H3(m, \delta, ID, PK_{ID}, h2)$.

3. Check whether $z\delta = h2g_{ID} + h3(y_{ID} + h1P_{pub})$ is valid, if so outputs "1", otherwise it outputs "0".

C. Intra-domain authentication phase.

The authentication of the device is performed by the edge computing node, and the authentication request is decentralized. The edge authentication node is close to the terminal to respond to the authentication request of the terminal in a timely manner.

a) $TD_A \rightarrow EA_A$:

$\{RS_{ID} || TD_{ID} || TD_{PK} || R2 || T || LT || Hash(Card_A) || Sig_{TD}(RS_{ID} || TD_{ID} || R2 || LT)\}$
The terminal TD_A sends authentication requests to the nearby edge authentication node EA_A , where $R2$ represents the intra-domain authentication request, T represents the timestamp, LT represents the lifecycle, and $Sig_{TD}(\cdot)$ represents the signature generated by the device TD_A invoking the signature algorithm using the private key, and $Hash(Card_A)$ represents the hash value of the identity credentials of the terminal TD_A .

b) The edge authentication node resolves the request $R2$ for intra-domain authentication, first check whether the RS_{ID} is the same domain, if not return an error message. If yes, then go to find out if TD_{ID} exists from the slave chain to determine whether the device is registered. If it does not exist, it means that the device is not registered and returns an error information. Otherwise, go to the next step.

c) Check if the device is within the lifecycle of the registration. If it is not, the identity credentials have expired, an error message is returned, and the device needs to be re-registered.

d) If within the lifecycle, the EA_A checks if the hash of the identity credentials sent by the device matches the hash of the identity credentials saved in the slave chain. If it matches, go to the next step, otherwise terminates the authentication process and returns an error message.

e) $EA_A \rightarrow TD_A : \{Result || EA_{PK} || Sig_{EA}(Result)\}$. EA_A uses the device public key to verify the signature of the device message to verify the authenticity of the message, signs the verification result and returns it to the device.

f) After receiving the signature of the authentication node, the device verifies the identity of the edge authentication node and the validity of the result by verifying the signature.

D. Cross-domain authentication phase.

a) $TD_A \rightarrow EA_B$:

$\{TD_{ID}||TD_{PK}||RS_{ID}||R3||T||params||Card_A||Sig_{TD}(TD_{ID}||RS_{ID}||R3||LT)\}$
 The device TD_A sends a cross-domain authentication request to the edge authentication node ES_B of domain B to access domain B, where R3 represents the cross-domain authentication request, T represents the timestamp, LT represents the life cycle, $Sig_{TD}(*)$ represents the signature generated by the device TD_A invoking the signature algorithm using the private key, $Card_A$ represents the identity credentials of the terminal TD_A , and params represents the public parameters of the domain where the device TD_A is located.

b) The EA_B receives the authentication request and analyzes it as a cross-domain request, then forwards the authentication request to the domain registry RS_B of the domain.

c) After receiving the request, the registration center RS_B first looks for the existence of RS_{ID} in the index information from the main chain, if it does not exist, it means that the domain where the device is located does not belong to the federated domain and has no right to access the B domain, and returns an error message to the edge authentication node. If it exists, then find whether TD_{ID} exists to determine whether the device is registered, if it does not exist, it means the device is not registered and returns an error message. Otherwise, go to the next step.

d) RS_B obtains the Merkle_Path of the slave chain where the device information is located according to the index information, verifies the validity of the device authentication information by SPV [20], and sends the verification result to the edge authentication node EA_B . If the authentication is successful, it goes to the next step. Otherwise, the edge authentication node directly returns the authentication failure message.

e) $EA_B \rightarrow TD_A: \{Result||EA_{PK}||Sig_{EA}(Result)\}$. The edge authentication node EA_B verifies whether the signature of the device is valid or not. If the signature verification is successful, it means that the authentication is successful and returns the signature of the verification result to the device. It also stores the identity information of the device into the slave chain of this domain so that the device can achieve fast authentication later.

f) The device receives the signature of the authentication node and verifies the identity of the edge authentication node and the validity of the result by verifying the signature.

V. ANALYSIS OF THE SECURITY AND EFFICIENCY OF THE SCHEME

A. Analysis of the Security

1) No Key escrow

The authentication process adopts a signature scheme based on no certificate, the private keys of the devices are generated by themselves, and KGC only generates part of the private keys, if the domain registry is controlled by the adversary or

evil, they do not know the complete private keys of the devices, so there is no risk of device private key leakage.

2) Ensure data integrity

The authentication process uses digital signature technology, where the sender signs the data sent with a private key and the verifier verifies the signature, which fails to be verified if the data ever changes. Secondly, the device registration information is written to the blockchain, which, as a distributed ledger, guarantees the integrity of the data due to its tamper-proof nature.

3) Non-repudiation

Each request is signed by the sender with its own private key, and the device cannot deny the authentication request it sends.

4) Mutual authentication

In the authentication process, the information of the device is stored in the block chain, and the identity of the device is authenticated by the smart contract. This scheme also requires the edge authentication node to sign the authentication result. The device can verify the signature to determine whether the result is trustworthy and the identity of the edge authentication node.

5) Against denial of service attack

In this paper, blockchain and edge computing technology are used to distribute authentication nodes at the edge end, and the blockchain acts as a distributed ledger to store device information, so that even if some nodes fail, the remaining nodes can still work normally and are not affected.

6) Against replay attack

In this paper, authentication requests with added timestamps are signed, and even if the request is intercepted by an attacker, the authentication node will refuse to serve the attacker by verifying the timeliness of the request by checking the signature.

7) Resist spoofing attacks

IoT devices are authenticated every time they communicate with the authentication nodes, and also verify the uniqueness of the identity by verifying the signature, the attacker cannot forge the identity of one of the parties to attack, so this scheme can resist spoofing attacks.

B. Analysis of Computation Overhead

Each entity is involved in a different cryptographic operation during the operation of the system. We summarised the cryptographic operations involved in the operation of the system and did not consider operations such as hashing, integer addition and multiplication as they took up very little time in the tests. The Pairing-Based Cryptography Library (PBC) was deployed on a local computer configured with an Intel (R) Core (TM) i5-5500U CPU with 8 GB of RAM. and the simulated experimental cryptographic operations are shown in Table I.

TABLE I. RUN TIME OF SEVERAL CRYPTOGRAPHIC OPERATIONS (IN MILLISECONDS).

Symbol	Description	Runtime(ms)
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Tsm	A scalar multiplication on elliptic curve	0.50
Tpsm	A pairing-based scalar multiplication	1.80
Tpa	A point addition on elliptic curve	0.02
Tbp	A bilinear pairing computation	6.00
Te	A exponentiation	1.00

This paper compares the authentication protocol based on identity signature [27] and the authentication protocol based on certificateless signature used in our scheme in terms of computational overhead. Table II shows the computational burden of the different components during the operation of the certificate-less signature (CLS) based authentication protocol and Table III shows the computational burden of the different components during the operation of the identity-based signature (IBS) based authentication protocol. From the data in the table, it is clear that the authentication protocol of this scheme has no bilinear pairwise and exponential operations, the cryptographic operation is relatively simple, and the computational overhead is better than that of the identity signature-based authentication protocol, which is suitable for IoT devices with limited computational power.

TABLE II. STATS ON TIME-CONSUMING CRYPTOGRAPHIC OPERATIONS OF CLS.

Phase	<i>TD</i>	<i>EA</i>	<i>RS</i>
Initialize and register	6Tsm+2Tpa	6Tsm+2Tpa	8Tsm+2Tpa
Intradomain authentication	5Tsm+2Tpa	5Tsm+2Tpa	/
Crossdomain authentication	5Tsm+2Tpa	5Tsm+2Tpa	5Tsm+2Tpa

TABLE III. STATS ON TIME-CONSUMING CRYPTOGRAPHIC OPERATIONS OF IBS.

Phase	<i>TD</i>	<i>EA</i>	<i>RS</i>
Initialize and register	2Tsm+Tpsm+ 3Tbp+2Te+Tpa	2Tsm+Tpsm+ 3Tbp+2Te+Tpa	2Tsm+Tpsm+ 3Tbp+2Te+Tpa
Intradomain authentication	2Tsm+Tpsm+ 3Tbp+2Te+Tpa	2Tsm+Tpsm+ 3Tbp+2Te+Tpa	/
Crossdomain authentication	2Tsm+Tpsm+ 3Tbp+2Te+Tpa	2Tsm+Tpsm+ 3Tbp+2Te+Tpa	2Tsm+Tpsm+ 3Tbp+2Te+Tpa

To further demonstrate the advantage of this scheme in terms of computational overhead, it was evaluated and compared with existing schemes [21-23]. Firstly, the computational overhead on the user side was compared, as

shown in Figure 2, and it can be seen that the performance of this scheme in terms of computational overhead on the user side is better than the other schemes. From Figure 3, it can be seen that the computational overhead on the server side of this scheme also outperforms other schemes.

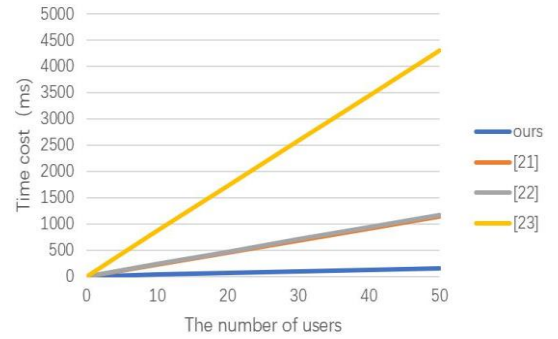


Figure 2. Cost on users for authentication

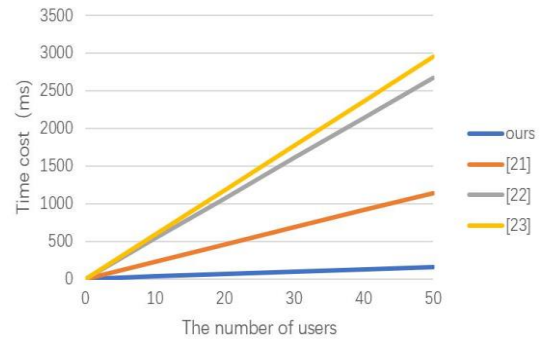


Figure 3. Cost on servers for authentication

C. Analysis of Storage Overhead

Unlike existing blockchain-based authentication schemes, in our scheme, the main chain only stores the index information of IoT devices, so this domain authentication node only needs to store the device information of the local domain, and additional simplified information is stored in the registry. Compared with the existing schemes [24-26] in which all nodes update the whole blockchain every time a new device is registered and each authentication node must store all registered device information of the local domain and other domains, the storage overhead of our scheme is much smaller and reduces the storage burden of the blockchain. Suppose there are 10 domains and each domain contains 1000 IoT devices. As described in this scheme, the length of authentication credentials, i.e., signature information, is about

40 bytes, while the length of index information block number is about 4 bytes. For the traditional blockchain-based authentication scheme, the storage overhead is about 40 KB, while for our proposed scheme, the storage overhead is about 4 KB + 4 KB (device information in local domain: 4 KB, index information in the main chain: 4 KB). As can be seen, this greatly reduces the storage space of the blockchain.

VI. CONCLUSION

The scheme addresses the drawbacks of traditional cross-domain authentication, and adopts blockchain technology and edge computing technology to design a main-slave blockchain cross-domain authentication architecture, which can realize distributed authentication and shorten authentication time. The scheme uses the slave chain to store the identity credentials of the device to achieve fast authentication within the device domain, and uses the main chain to build trust between different domains, stores index information, and shares the device authentication information of different domains to achieve cross-domain authentication of the device. The authentication request is decentralized using edge computing technology to reduce the authentication time and network latency. In addition, the authentication process adopts an identity authentication protocol based on certificateless signature, allowing devices to generate their own public and private keys, eliminating problems such as key escrow based on identity signature and certificate management based on digital certificates, and reducing storage overhead and computation costs. Finally, by analyzing the security and efficiency of the certificateless signature-based authentication protocol, it is proved that this scheme has better security and efficiency. In the next work, we will further study the cross-chain technology, open up the chain data channel for cross-domain authentication, realize the trusted interaction between blockchains, and further improve the processing efficiency of cross-domain authentication in blockchains.

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